

HODGE THEORY

The $\bar{\partial}$ -Poincaré lemma

One complex dimension

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The equation we want to solve

Let $D \subset \mathbb{C}$ be a disk, with coordinate $z = x + iy$.

Given a smooth $(0, 1)$ -form

$$\eta = g(z, \bar{z}) d\bar{z}$$

extending smoothly to a neighbourhood of $\bar{D}$, we want

$$\bar{\partial}u = \eta, \quad u \in C^\infty(D).$$

In coordinates this is just

$$\bar{\partial}u = \frac{1}{2}(u_x + iu_y) d\bar{z}.$$

We first solve the equation on a torus up to a constant term.

Hodge theory

Functions on the square torus

We start with the elliptic curve

$$E = \mathbb{C}/(2\pi\mathbb{Z} + 2\pi i\mathbb{Z}).$$

Think of $(0, 2\pi] \times (0, 2\pi]$ with opposite edges identified.

A function on E is a function on $\mathbb{C}$ satisfying

$$f(x + 2\pi, y) = f(x, y), \quad f(x, y + 2\pi) = f(x, y).$$

There are two real coordinates, so there will be two frequencies.

The Fourier basis

The functions

$$e_{n,m}(x, y) = e^{i(nx+my)} = e^{inx} e^{imy}, \quad (n, m) \in \mathbb{Z}^2,$$

form an orthonormal basis of $L^2(E)$ for the measure $dx dy / (2\pi)^2$.

Therefore every $f \in L^2(E)$ has a basis expansion

$$f = \sum_{n,m \in \mathbb{Z}} a_{n,m} e_{n,m}, \quad a_{n,m} = \langle f, e_{n,m} \rangle.$$

The equality means convergence in L^2 . We use the Fourier basis theorem without proving it here.

Fourier modes of a form

Expand the coefficient of a smooth $(0, 1)$ -form:

$$\eta = g d\bar{z} = \sum_{n,m \in \mathbb{Z}} \underbrace{a_{n,m} e_{n,m}}_{\eta_{n,m}} d\bar{z}.$$

The summand $\eta_{n,m}$ is its (n, m) -Fourier mode.

$$\eta = a_{0,0} d\bar{z} + \sum_{(n,m) \neq (0,0)} \eta_{n,m}.$$

The zero mode has constant coefficients. A nonzero frequency means $(n, m) \neq (0, 0)$.

Differentiation preserves the frequency

Differentiate one basis function:

$$de_{n,m} = e_{n,m}(in\,dx + im\,dy),$$

$$\bar{\partial}e_{n,m} = \frac{in - m}{2} e_{n,m} d\bar{z}.$$

Every coefficient is a constant multiple of the same $e_{n,m}$.

Thus d and $\bar{\partial}$ preserve the pair (n, m) . We can solve the equation one frequency at a time.

Closed forms and exact forms

A form η is $\bar{\partial}$ -closed if $\bar{\partial}\eta = 0$.

It is $\bar{\partial}$ -exact if $\eta = \bar{\partial}u$ for some u .

In complex dimension one, every $(0, 1)$ -form is closed:

$$\bar{\partial}(g d\bar{z}) = \frac{\partial g}{\partial \bar{z}} \underbrace{d\bar{z} \wedge d\bar{z}}_0 = 0.$$

Our task is to find primitives. Closedness is already automatic in this degree.

The complex Cartan formula

Put $X = \partial_x$, so $IX = \partial_y$.

For these constant vector fields,

$$\bar{\partial}\iota_X + \iota_X\bar{\partial} = \frac{1}{2}(\mathcal{L}_X + i\mathcal{L}_{IX}).$$

Here ι_X is contraction. On a one-form, $\iota_X\eta = \eta(X)$. In particular,

$$\iota_X(g d\bar{z}) = g.$$

Here $\mathcal{L}_X$ and $\mathcal{L}_{IX}$ simply differentiate the coefficients in the x - and y -directions.

Primitives of nonzero modes

For $\eta_{n,m} = a_{n,m} e_{n,m} d\bar{z}$, we have

$$\mathcal{L}_X \eta_{n,m} = in \eta_{n,m}, \quad \mathcal{L}_{\bar{X}} \eta_{n,m} = im \eta_{n,m}.$$

Since $\bar{\partial} \eta_{n,m} = 0$, Cartan's formula gives

$$\bar{\partial}(\iota_X \eta_{n,m}) = \frac{in - m}{2} \eta_{n,m}.$$

If $(n, m) \neq (0, 0)$, then $in - m \neq 0$, hence

$$\eta_{n,m} = \bar{\partial} \left(\frac{2a_{n,m}}{in - m} e_{n,m} \right).$$

Adding the primitives

Define a function on the torus by

$$f = \sum_{(n,m) \neq (0,0)} \frac{2a_{n,m}}{in - m} e_{n,m}.$$

Differentiating term by term gives

$$\bar{\partial}f = \eta - a_{0,0} d\bar{z}.$$

Therefore

$$\eta = c d\bar{z} + \bar{\partial}f, \quad c = a_{0,0}.$$

Fourier fact used here: for smooth g , this series and all its derivatives converge uniformly, so f is smooth.

Hodge theory

The constant part in cohomology

Dolbeault cohomology identifies closed forms that differ by an exact form:

$$H_{\bar{\partial}}^{0,1}(E) = \frac{\{\text{smooth closed } (0,1)\text{-forms}\}}{\{\bar{\partial}f : f \in C^\infty(E)\}}.$$

Since $\eta = c d\bar{z} + \bar{\partial}f$,

$$[\eta] = c[d\bar{z}], \quad H_{\bar{\partial}}^{0,1}(E) \cong \mathbb{C}.$$

$\bar{\partial}f$ always has zero constant part: differentiation preserves frequencies and kills the constant term. Thus $c d\bar{z}$ is exact only when $c = 0$.

The disk inside an elliptic curve

Choose $L > 0$ so large that $\overline{D}$ lies in the interior of a fundamental square for

$$E_L = \mathbb{C}/(L\mathbb{Z} + iL\mathbb{Z}).$$

The quotient map embeds a neighbourhood of $\overline{D}$ into E_L .

The same Fourier calculation works for any period L .

Rescaling the coordinate by $w = 2\pi z/L$ reduces it to the square torus we just used.

On this neighbourhood, z and $\bar{z}$ are well-defined functions.

Extending the form to the torus

Let U be a neighbourhood of $\overline{D}$ inside that square, where η is defined.

Choose a smooth function χ that is 1 near $\overline{D}$ and has support contained in U .

$$\tilde{\eta} = \begin{cases} \chi\eta & \text{on } U, \\ 0 & \text{outside } U. \end{cases}$$

This is a smooth $(0,1)$ -form on E_L , equal to η on D .

It is automatically $\bar{\partial}$ -closed, just like every $(0,1)$ -form on a complex curve.

The primitive on the disk

On the elliptic curve, our calculation gives

$$\tilde{\eta} = c d\bar{z} + \bar{\partial}f.$$

On the disk, $\bar{z}$ is a function and $\bar{\partial}\bar{z} = d\bar{z}$. Restricting to D therefore gives

$$\eta = \bar{\partial}(c\bar{z} + f|_D).$$

This is the required primitive.

On the whole torus, $\bar{z}$ is not periodic, so it cannot serve as a global primitive.

Supplement: the $(1, 1)$ case

A $(1, 1)$ -form on the disk is $\eta = g \, dz \wedge d\bar{z}$.

By the $(0, 1)$ case, choose u with

$$\bar{\partial}u = g \, d\bar{z}.$$

Then

$$\bar{\partial}(-u \, dz) = -\bar{\partial}u \wedge dz = g \, dz \wedge d\bar{z} = \eta.$$

These are the two possible types with positive antiholomorphic degree in complex dimension one: $(0, 1)$ and $(1, 1)$.

Supplement: the constant representatives

The same calculation for each type gives

Type	Constant representative	Cohomology on E
$(0, 0)$	1	$H_{\bar{\partial}}^{0,0}(E) \cong \mathbb{C}$
$(1, 0)$	dz	$H_{\bar{\partial}}^{1,0}(E) \cong \mathbb{C}$
$(0, 1)$	$d\bar{z}$	$H_{\bar{\partial}}^{0,1}(E) \cong \mathbb{C}$
$(1, 1)$	$dz \wedge d\bar{z}$	$H_{\bar{\partial}}^{1,1}(E) \cong \mathbb{C}$

Here (p, q) records the number of dz and $d\bar{z}$ factors. Each cohomology class has a unique constant representative.